

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

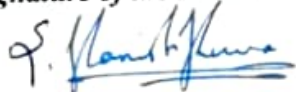
Descriptive Written Test

Code of Conduct:

(192511415)

I Kanish Kumar S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:





Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

ASSESSMENT TOOL - 3

① SMART HOME AUTOMATION SYSTEM USING INTELLIGENT AGENTS :

A smart home automation system controls lighting, temperature and security using sensors such as motion detectors, temperature sensors and time-based inputs. It also learns user preferences to improve comfort and energy efficiency.

TYPES OF INTELLIGENT AGENTS :

1. Simple Reflex Agent :

- * Works only on the current input using condition-action rules.
- * Does not remember previous states.

Example :

- If motion is detected → Turn ON lights
- If no motion is detected → Turn OFF lights
- * ADVANTAGE → Fast and Simple.
- * DISADVANTAGE → Cannot learn or adapt.

2. MODEL BASED AGENT :

- * Maintains an internal model of the environment.
- * Uses previous information along with current inputs.
- * ADVANTAGE → Handles partially observable environments.
- * DISADVANTAGE → More complex than reflex agents.

3. GOAL - BASED AGENT :

- * Makes decisions to achieve specific goals.
- * Goals include,

- Maintaining comfortable temperature
- Saving electricity
- Ensuring home security

4. UTILITY BASED AGENT:

- * Selects the action that provides maximum benefit.
- * Considers multiple factors such as comfort, energy consumption and security.

* EXAMPLE:

→ Reduces AC Usage to save electricity while keeping the room comfortable.

- * ADVANTAGE → Produces the best overall performance.

ANALYSIS:

- * Simple reflex Agent is fast but cannot learn.
- * Model-Based Agent remembers previous states.
- * Goal-Based Agent focuses on achieving objectives.
- * Utility-Based Agent chooses the most beneficial action.

CONCLUSION:

A Hybrid Agent (Model-Based + Goal-Based + Utility-Based) is the best solution because it learns used behaviour, responds to real-time sensor inputs, minimizes energy consumption and improves comfort and security.

ROBOT IN AN UNKNOWN MAZE USING UNINFORMED SEARCH:

A robot is placed in an unknown maze and must find the exit without prior knowledge of the environment. Uninformed search algorithms help the robot explore the maze systematically.

UNIFORM COST SEARCH (UCS):

- Expands the node with the lowest cumulative path cost.
- Finds the least-cost (optimal) path.
- Suitable when edge costs are different.

ADVANTAGES:

- * Complete
- * optimal
- * Finds the cheapest path.

DISADVANTAGES:

- * High memory usage.
- * slower for large search spaces.

ITERATIVE DEEPENING SEARCH (IDS):

- Combines the advantages of DFS and BFS.
- Repeatedly performs depth-limited searches with increasing depth.
- Useful when the goal depth is unknown.

ADVANTAGES:

- * Complete
- * Requires less memory

* Suitable for deep searches.

DISADVANTAGES:

- Repeats search at every depth level.
- More computation time.

TIME AND SPACE COMPLEXITY:

ALGORITHM	TIME COMPLEXITY	SPACE COMPLEXITY
UCS IDS	Exponential $O(b^d)$	Exponential $O(bd)$

RELATION TO INTELLIGENT AGENTS:

- Simple Reflex Agent : Cannot solve unknown mazes effectively.
- Model-Based Agent : Builds a map of explored areas.
- Goal-Based Agent : Plans actions to reach the exit.
- Utility-Based Agent : chooses the safest or least-cost path.

CONCLUSION:

Goal-Based Agents combined with UCS provide accurate and optimal navigation, making them ideal for robots exploring unknown environments.

RUBRIC FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand